

Actuator Optimal Design

Lecture 1 : Pareto Optimality & Design Principles

2026/03/02

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Pareto Optimality

좋은 모터는 어떻게 만들 수 있는가?

라는 질문을 던지기 전에...

무엇이 더 ‘좋은’ 자전거인가?



VS



무엇이 더 ‘좋은’ 자전거인가?



VS



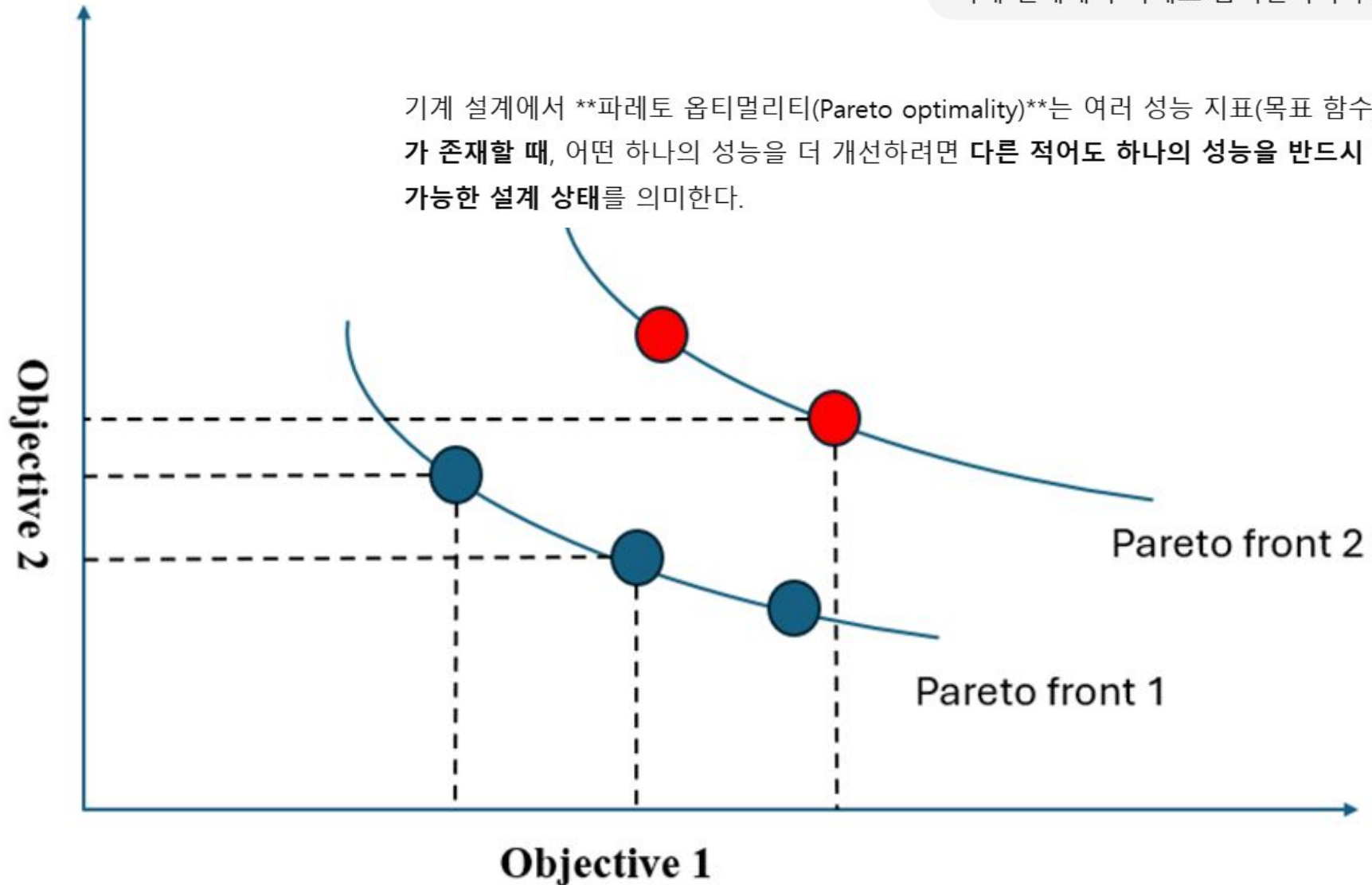
무엇이 **좋은** 자전거 / 모터 / 설계인가?

좋은은 어떻게 정의할 수 있는가?

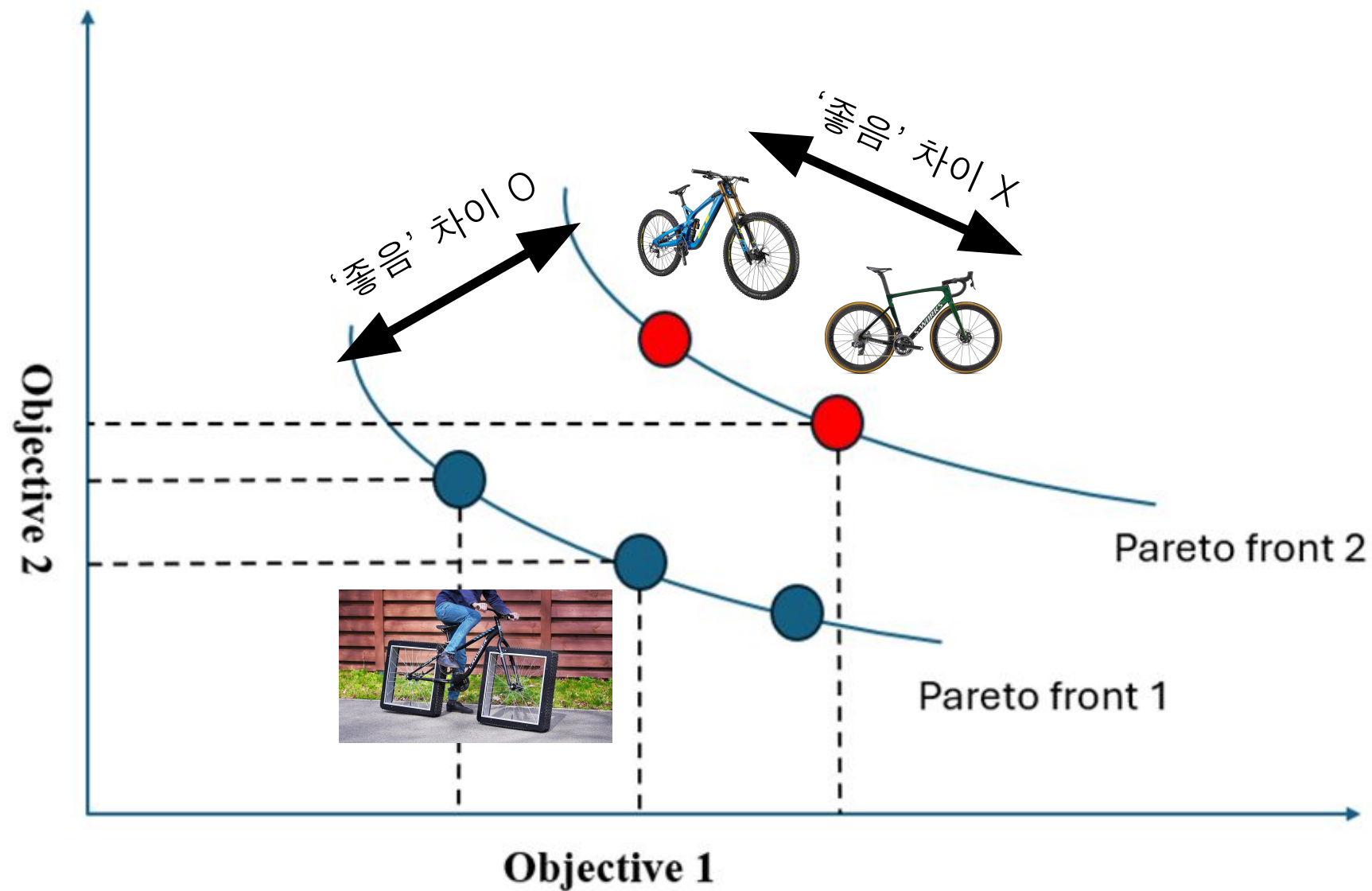
Concept of Pareto Optimality

기계 설계에서 파레토 옵티멀리티가 의미하는 바가 뭐지?

기계 설계에서 ****파레토 옵티멀리티(Pareto optimality)****는 여러 성능 지표(목표 함수) 사이에 **트레이드오프**가 존재할 때, 어떤 하나의 성능을 더 개선하려면 다른 적어도 하나의 성능을 반드시 악화시키지 않고는 불가능한 설계 상태를 의미한다.



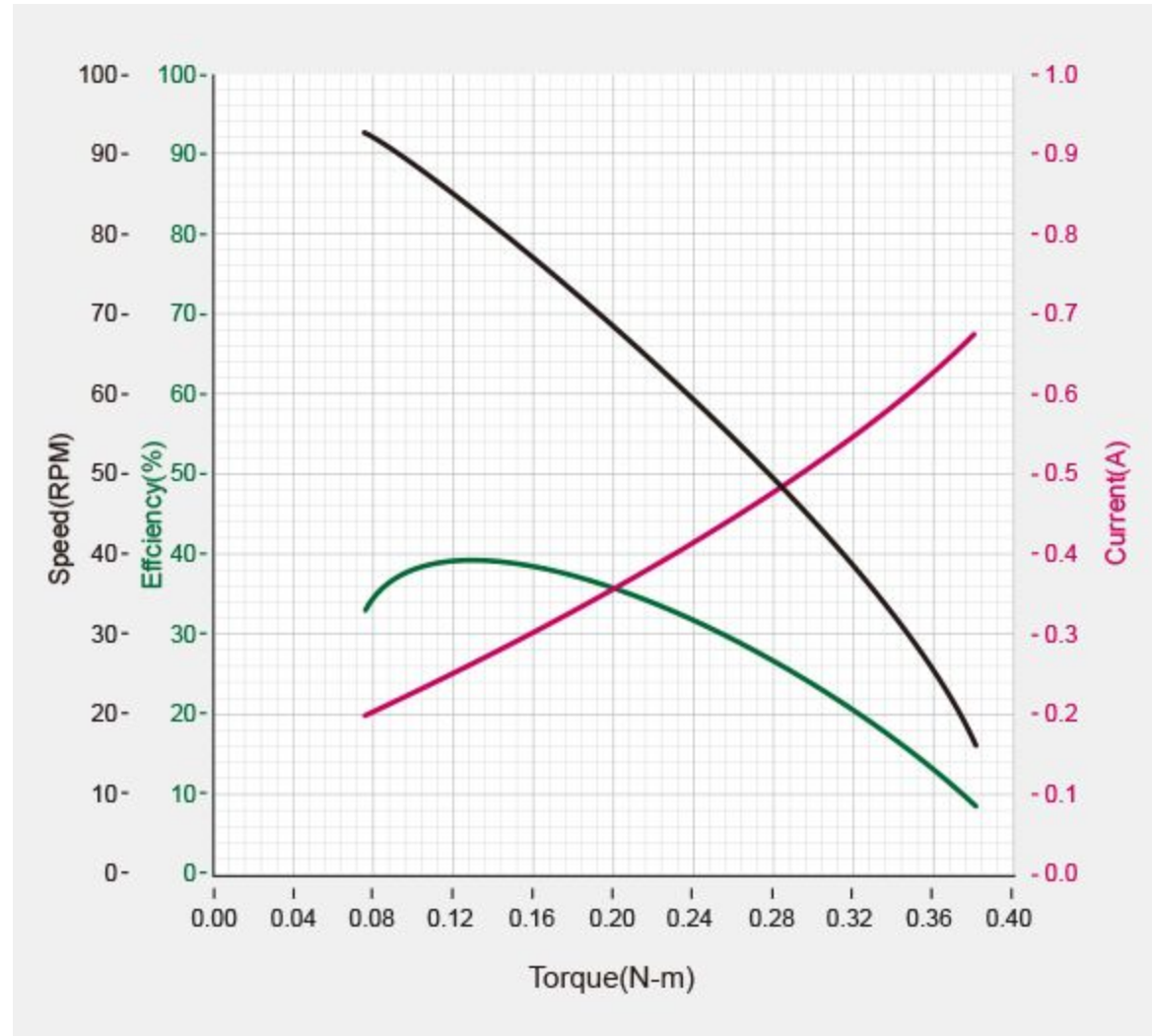
Concept of Pareto Optimality



그렇다면 모터의 좋음을 정의하는
Metric들은 무엇인가?

Is Power a Metric for Motors?

10



Is Power a Metric for Motors?

11



KV	1980KV
LiPo Cells (LiPo)	6S
No-Load Current	3.2A/24V
Maximum current (Non-hermetic Ambient Temperature $\leq 25^{\circ}\text{C}$)	49A/12s
Rated Power Output	150W
Max Power	1166W/12s
Maximum Thrust	1868g(HQ-5037)
Motor Weight	38g (With wire)
Motor Size	$\Phi 28 \times H19.9\text{mm}$
Slot/Pole	12N14P
Wire Size	20AWG

Is Power a Metric for Motors?



LiPo Cells : 24S

Motor weight : Stator + Rotor : **1042g**

Max Power : **2700W**

Is Power a Metric for Motors?

13



Mass : 38g

Power : 1166W

VS



Mass : 1042g (27x)

Power : 2700W

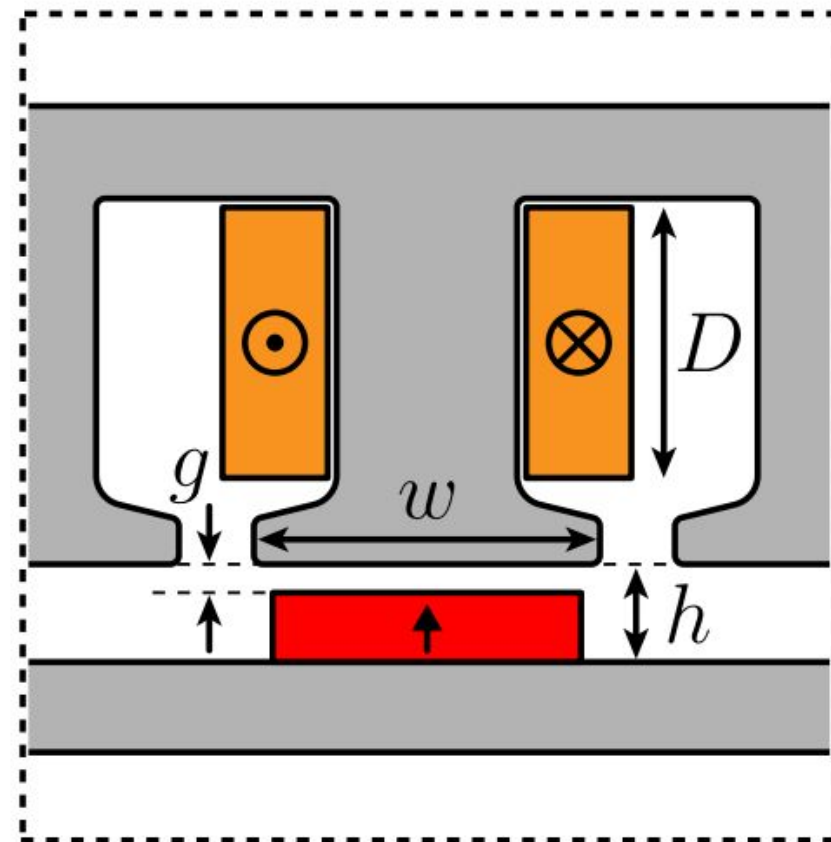
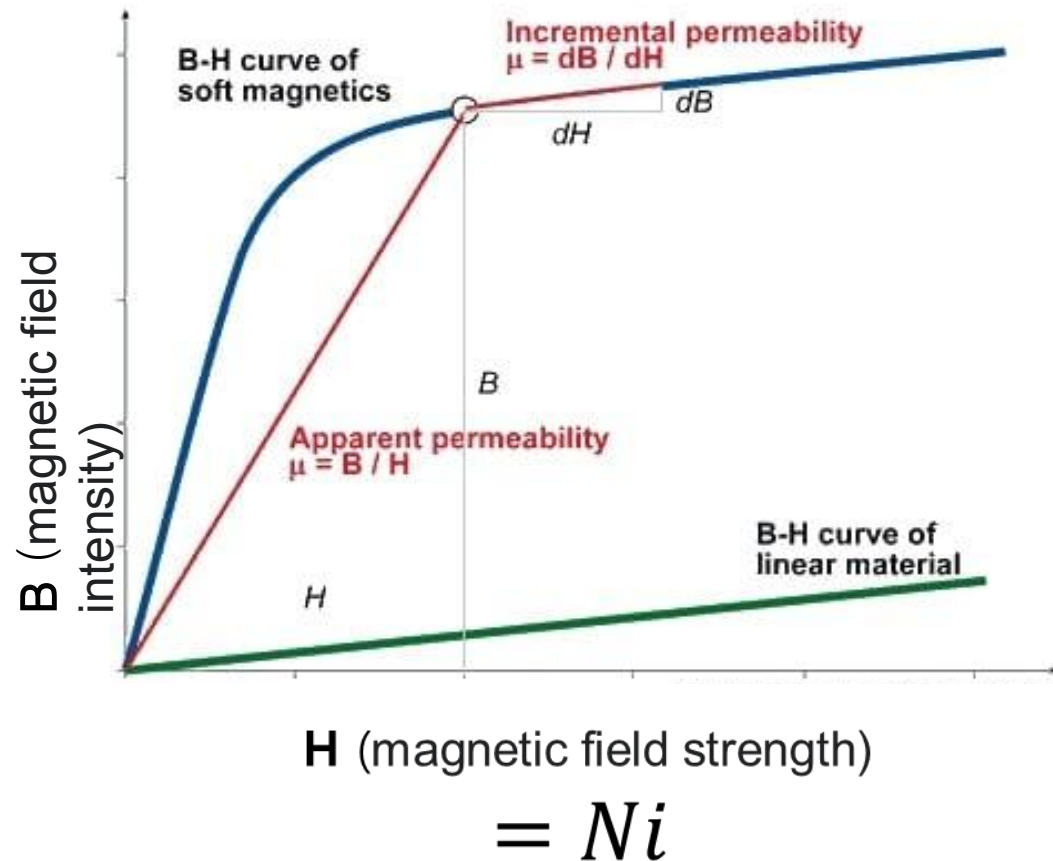
(2.3x)

Motor Design Principles

1. Torque Characteristic is invariant with Winding

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- 고정된 stator-rotor dimension에 대해, 모터의 토크 특성은 Winding(turns, diameter)과 무관하다.

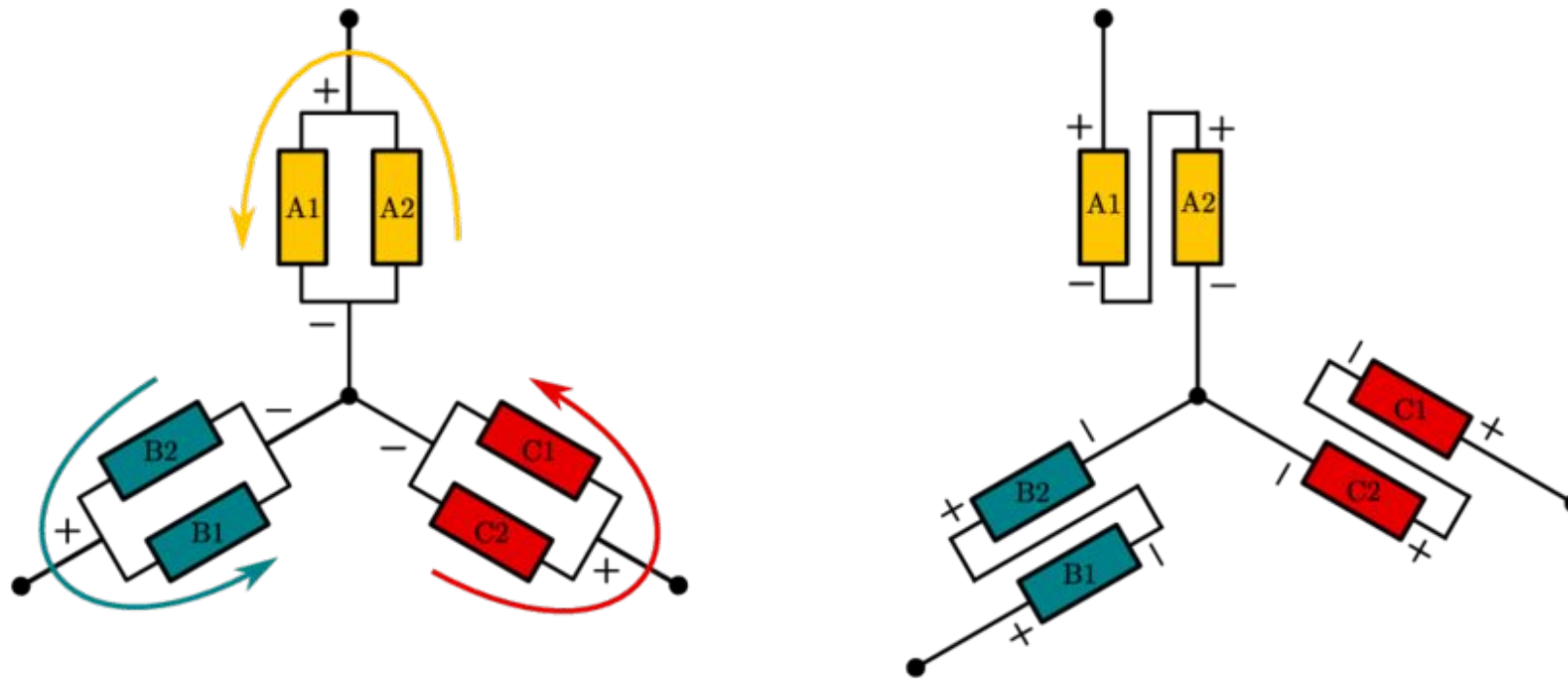


2. Winding is related to Motor Driver

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- 최적의 winding (delta/wye, serial/parallel, coil diameter/turns)은

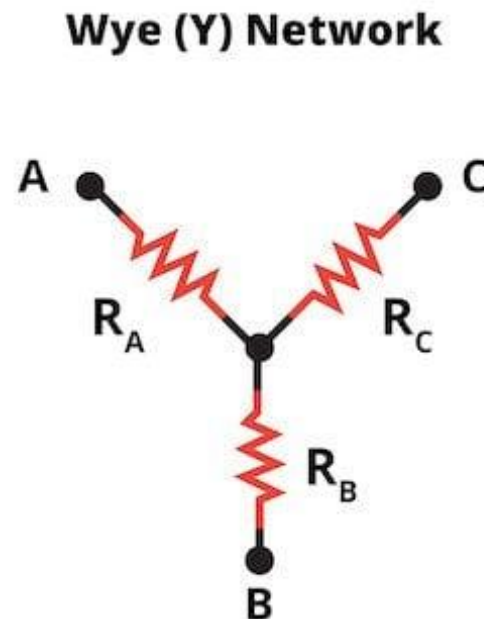
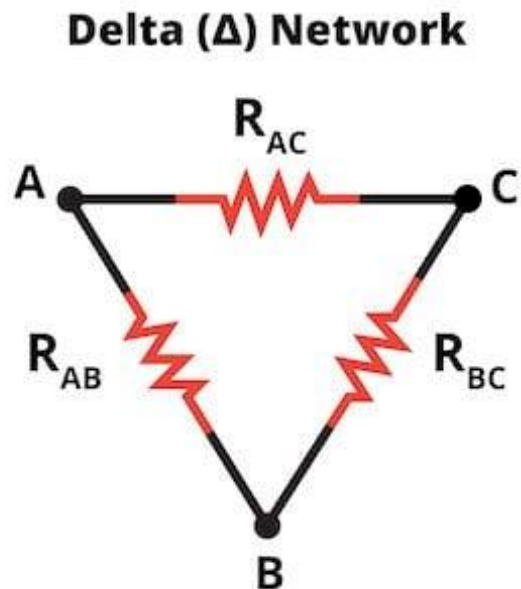
Motor driver의 최대 line voltage/line current 출력 성능에 의해 결정된다.



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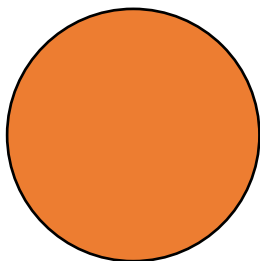
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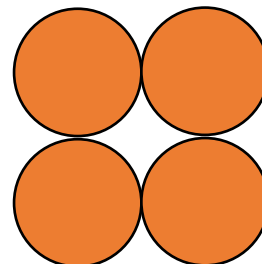
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Low turns,
Big diameter



High turns,
small diameter

3. With fixed Mass & Torque, Inertia is fixed.

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Mass : 1kg

Reduction ratio : 100

Torque(rotor) : 1Nm

Torque(joint) : 100Nm



Mass : 1kg

Reduction ratio : 10

Torque(rotor) : 10Nm

Torque(joint) : 100Nm

3. With fixed Mass & Torque, Inertia is fixed.

- Rotor-stator 질량과 joint max torque(혹은 joint 기준 Motor Constant: K_m)를 고정할 때, joint inertia는 stator-rotor dimension과 감속비에 무관하게 결정된다.

viscous, and frictional mechanical impedance). Suppose we want to choose a motor and gearset with the largest torque per inertia while keeping the output torque and motor mass constant. If we double the radius, the motor length is halved, the motor torque doubles and the gear ratio is halved. The motor rotor inertia increases by 4 times but the reflected rotor inertia through the gears⁴ scales by one fourth. **The total reflected rotor inertia and the total torque at the output shaft stay the same.** In general, higher gear ratios add mass, inertia and friction loss. Thus the increased motor radius would

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Torque

Mass

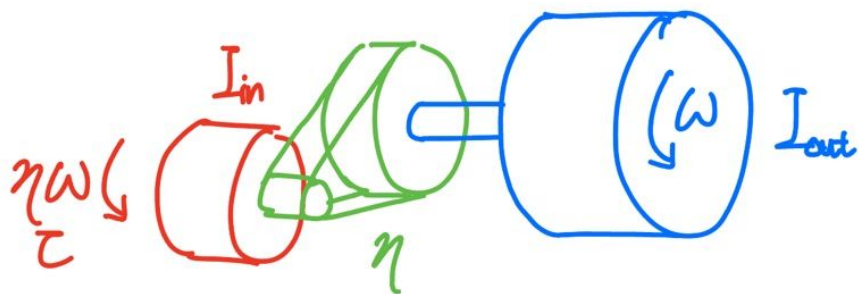
Inertia

Any Questions?

Appendix

+Optimal reduction ratio?

- Output(joint) 각가속도를 최대화하기 위한 I_{out} 과 I_{in} 의 비율: $I_{out} = \eta^2 I_{in}$



$$(\tau - I_{in} \eta \dot{\omega}) \eta = I_{out} \dot{\omega}$$

$$\eta \tau = (I_{out} + \eta^2 I_{in}) \dot{\omega}$$

$$\dot{\omega} = \left(\frac{\eta}{I_{out} + \eta^2 I_{in}} \right) \tau$$

$$\frac{d}{d\eta} \left(\frac{\eta}{I_{out} + \eta^2 I_{in}} \right) = (I_{out} + \eta^2 I_{in})^{-1} - 2\eta^2 (I_{out} + \eta^2 I_{in})^{-2} I_{in} = 0$$

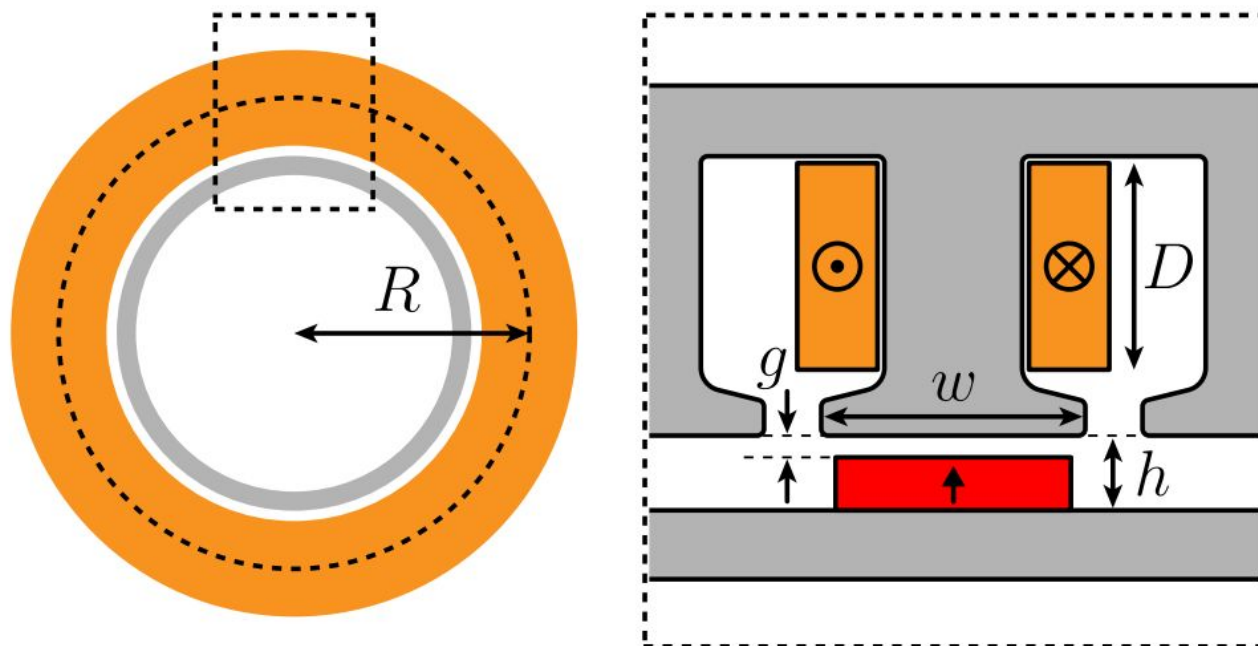
$$I_{out} = \eta^2 I_{in}$$

Basic Design Parameters

- Stator-Rotor dimensions: gap radius, thickness, length
- Reducer: reduction ratio
- Winding: delta/wye, serial/parallel, coil diameter/turns

Design Parameters

- Stator-Rotor dimensions: gap radius, thickness, length
- Reducer: reducer type, reduction ratio
- Winding: delta/wye, serial/parallel, coil diameter/turns

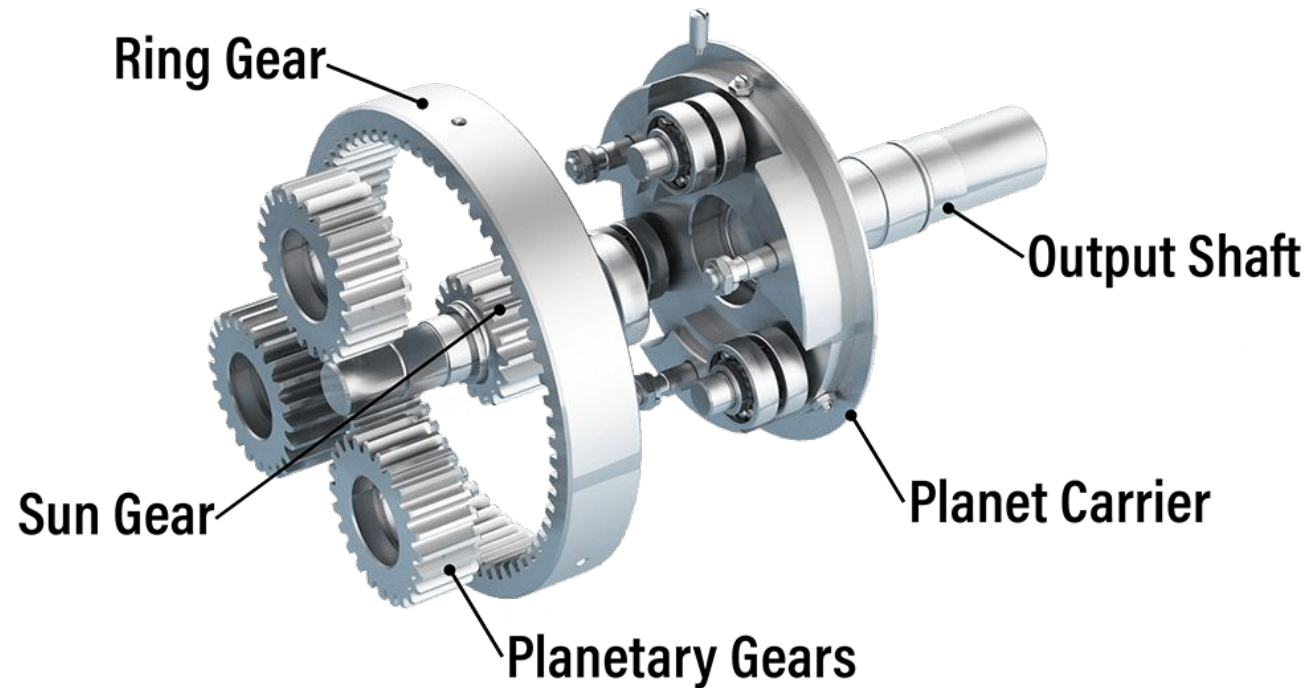


R for gap radius,
 l for axial length,
 D for thickness,
 P for # of poles.

Design Parameters

30

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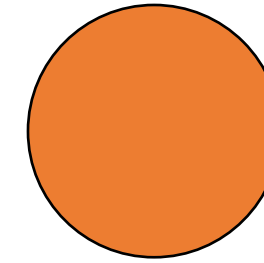


η_r for reduction ratio.

Design Parameters

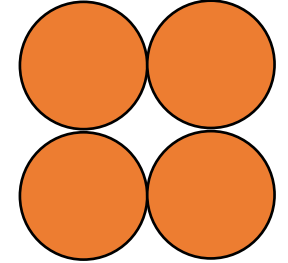
31

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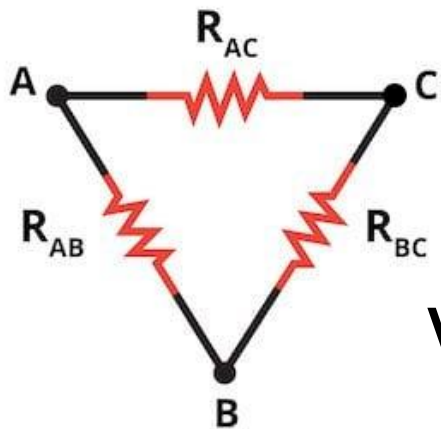
Low turns,
Big diameter

VS

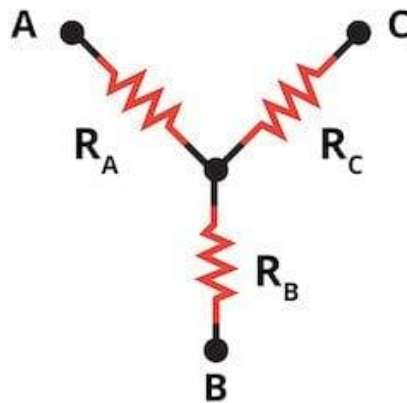


High turns,
small diameter

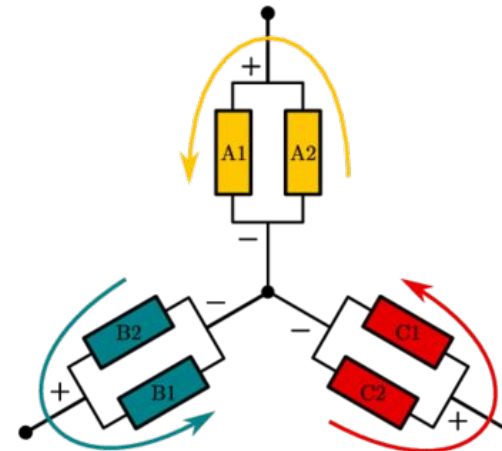
Delta (Δ) Network



Wye (Y) Network



VS



VS

